

UNIVERSIDAD AUTONOMA DE NUEVO LEON



Facultad de Ciencias Físico Matemáticas

(FCFM)

Unidad de Aprendizaje:

Desarrollo web – tecnologías servidor

Actividad

Nombre: Andrés Castro Hernandez

Matricula: 2069855

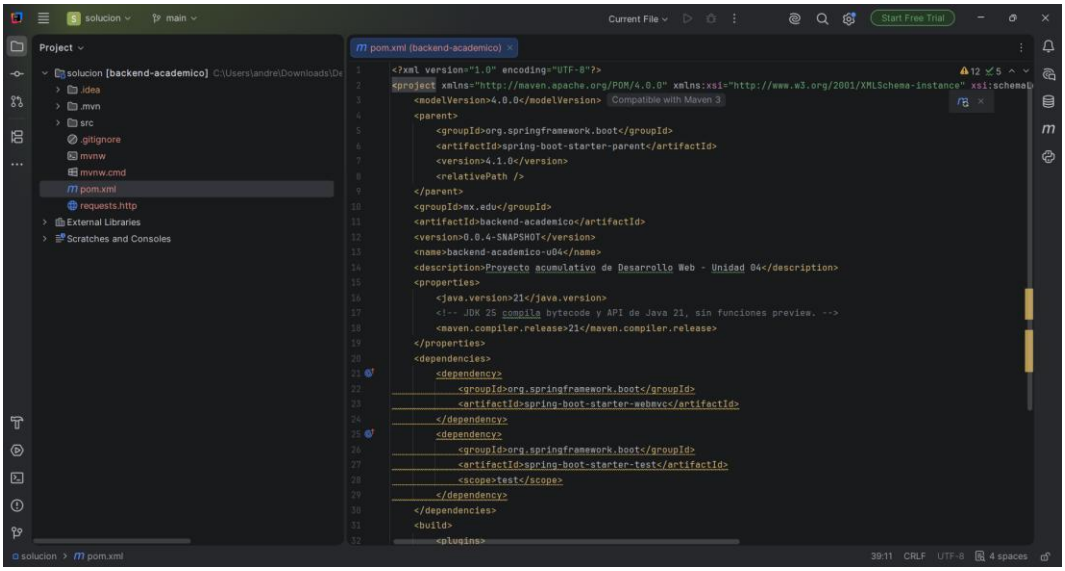
Grupo: 031

Profesor: Gustavo Adolfo Rodríguez Martínez

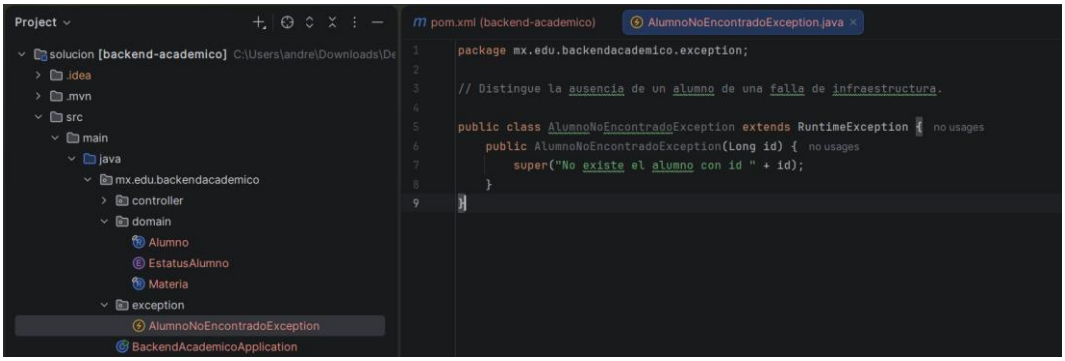
Carrera: LCC

Monterrey N.L. a; 16 de octubre de 2026

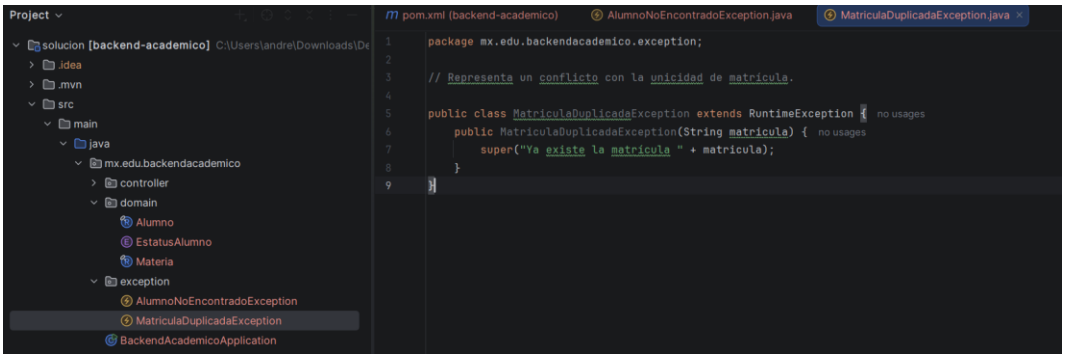
Acción: modificar pom.xml.



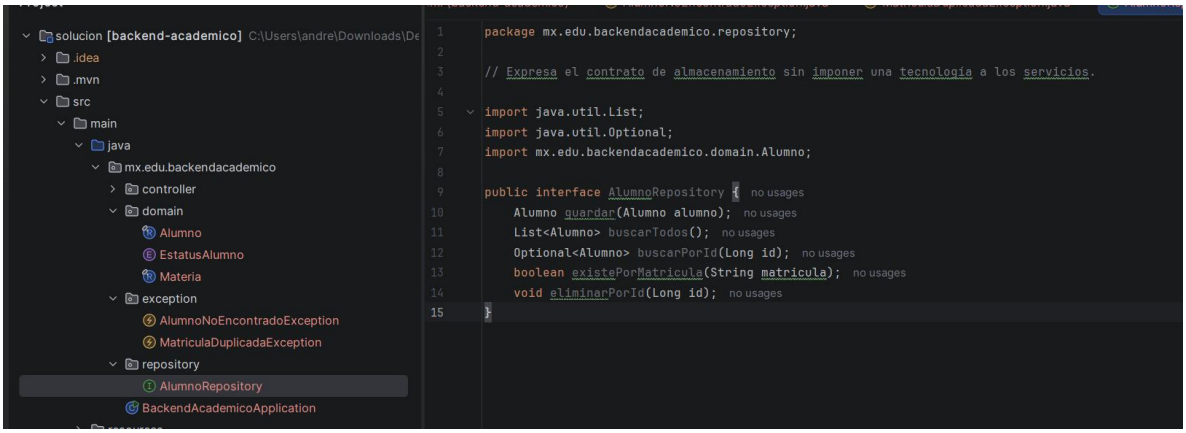
Acción: crear src/main/java/mx/edu/backendacademico/exception/AlumnoNoEncontrado Exception.java.



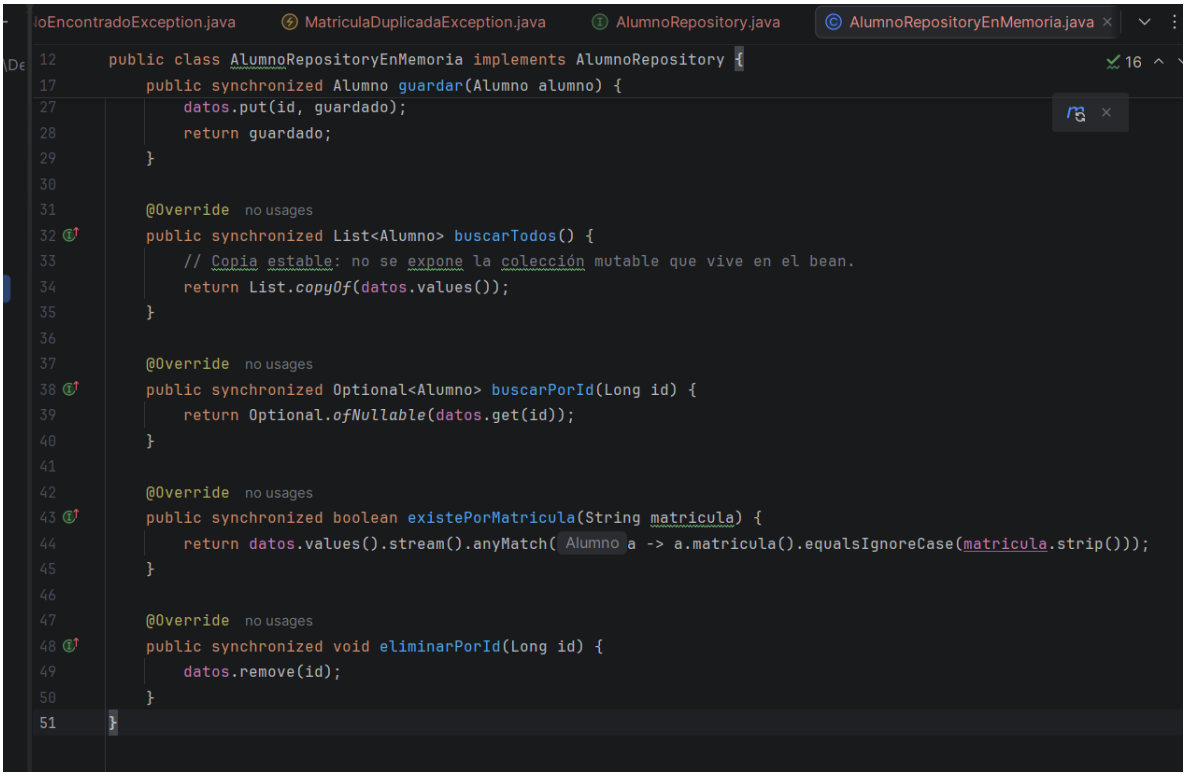
Acción: crear src/main/java/mx/edu/backendacademico/exception/MatriculaDuplicadaEx ception.java.



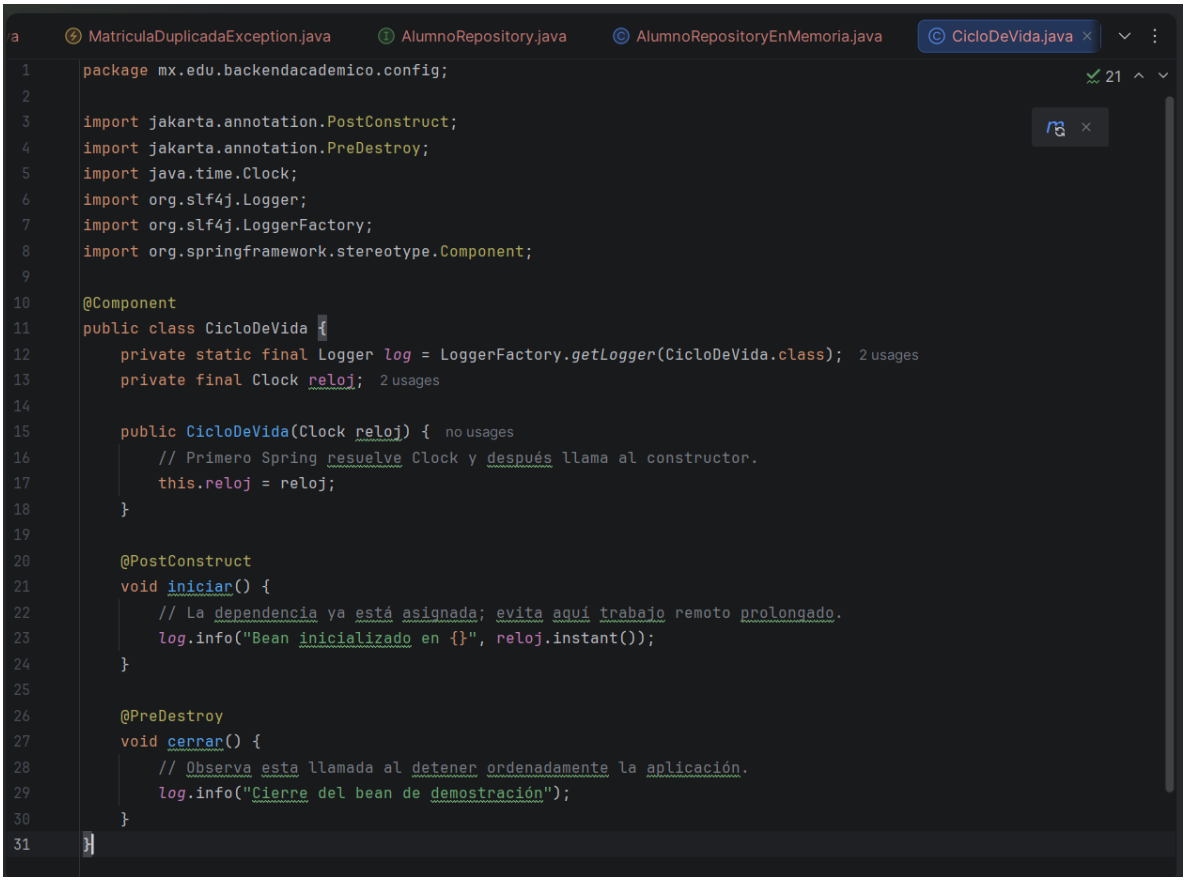
Acción: crear src/main/java/mx/edu/backendacademico/repository/AlumnoRepository.java.



Acción: crear src/main/java/mx/edu/backendacademico/repository/AlumnoRepositoryEnMemoria.java.



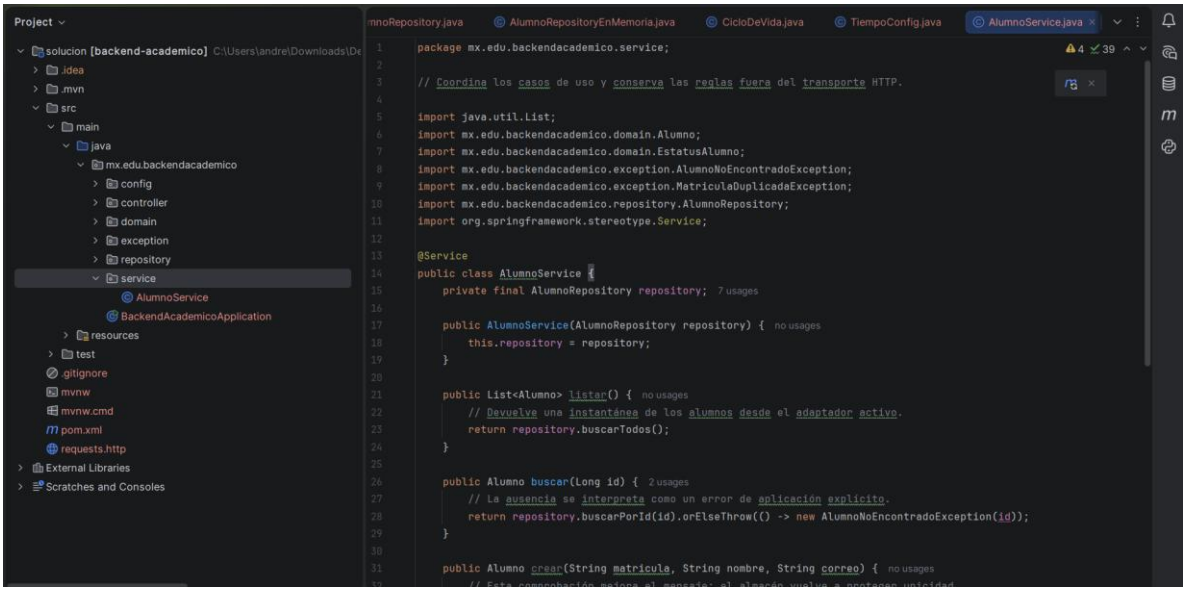
Acción: crear src/main/java/mx/edu/backendacademico/config/CicloDeVida.java.



Acción: crear src/main/java/mx/edu/backendacademico/config/TiempoConfig.java.



Acción: crear src/main/java/mx/edu/backendacademico/service/AlumnoService.java.



Comprobación del resultado

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 04\solucion> .\mvnw.cmd
test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u04 0.0.4-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 12 source files with javac [debug parameters release 21] to target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 04\solucion\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 2 source files with javac [debug parameters release 21] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.backendacademico.BackendAcademicoApplicationTests
23:32:47.472 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils --
Could not detect default configuration classes for test class
```

```
[mx.edu.backendacademico.BackendAcademicoApplicationTests]: BackendAcademicoApplicationTests does not
declare any static, non-private, non-final, nested classes annotated with @Configuration.
23:32:47.767 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found
@SpringBootConfiguration mx.edu.backendacademico.BackendAcademicoApplication for test class
mx.edu.backendacademico.BackendAcademicoApplicationTests
23:32:47.919 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils --
Could not detect default configuration classes for test class
[mx.edu.backendacademico.BackendAcademicoApplicationTests]: BackendAcademicoApplicationTests does not
declare any static, non-private, non-final, nested classes annotated with @Configuration.
23:32:47.921 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found
@SpringBootConfiguration mx.edu.backendacademico.BackendAcademicoApplication for test class
mx.edu.backendacademico.BackendAcademicoApplicationTests

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v4.1.0)

2026-10-16T23:32:48.398-06:00  INFO 24780 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplicationTests   : Starting BackendAcademicoApplicationTests using Java 24.0.2
with PID 24780 (started by andre in C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 04\solucion)
2026-10-16T23:32:48.400-06:00  INFO 24780 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplicationTests   : No active profile set, falling back to 1 default profile:
"default"
2026-10-16T23:32:49.355-06:00  INFO 24780 --- [backend-academico] [           main]
m.e.backendacademico.config.CicloDeVida  : Bean inicializado en 2026-10-17T05:32:49.355370300Z
2026-10-16T23:32:49.917-06:00  INFO 24780 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplicationTests   : Started BackendAcademicoApplicationTests in 1.897 seconds
(process running for 3.525)
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future
releases of the JDK. Please add Mockito as an agent to your build as described in Mockito's
documentation: https://javadoc.io/doc/org.mockito/mockito-
core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\andre\.m2\repository\net\bytebuddy\byte-
buddy-agent\1.18.10\byte-buddy-agent-1.18.10.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this
warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more
information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because
bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.611 s -- in
mx.edu.backendacademico.BackendAcademicoApplicationTests
[INFO] Running mx.edu.backendacademico.domain.AlumnoTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.011 s -- in
mx.edu.backendacademico.domain.AlumnoTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.313 s
[INFO] Finished at: 2026-10-16T23:32:50-06:00
[INFO] ------------------------------------------------------------------------
------
```

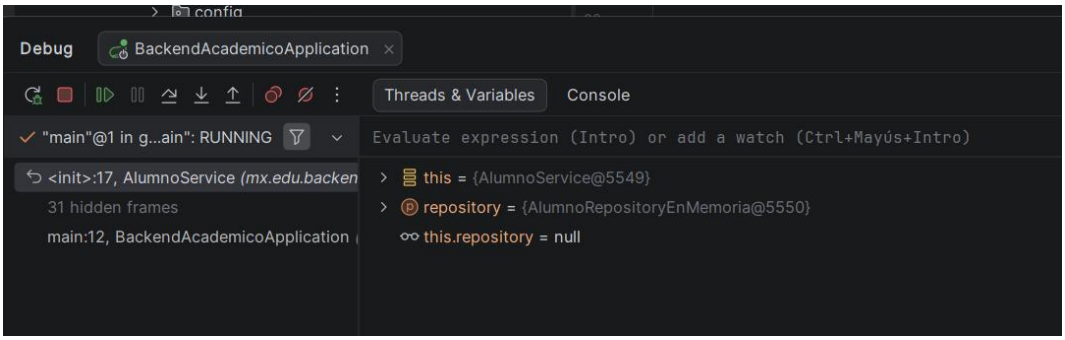
```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad 04\solucion> .\mvnw.cmd
spring-boot:run
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u04 0.0.4-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ backend-academico >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 04\solucion\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ backend-academico <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ backend-academico ---
[INFO] Attaching agents: []

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v4.1.0)

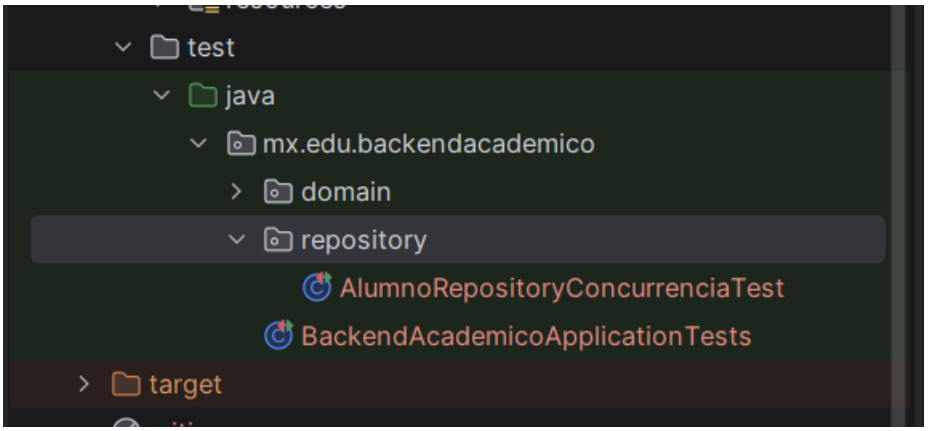
2026-10-16T23:34:09.908-06:00  INFO 19520 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : Starting BackendAcademicoApplication using Java 24.0.2 with
PID 19520 (C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad
04\solucion\target\classes started by andre in C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-
servidor\Unidades\Unidad 04\solucion)
2026-10-16T23:34:09.912-06:00  INFO 19520 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : No active profile set, falling back to 1 default profile:
"default"
2026-10-16T23:34:10.624-06:00  INFO 19520 --- [backend-academico] [           main]
o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8080 (http)
2026-10-16T23:34:10.638-06:00  INFO 19520 --- [backend-academico] [           main]
o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-10-16T23:34:10.639-06:00  INFO 19520 --- [backend-academico] [           main]
o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-10-16T23:34:10.696-06:00  INFO 19520 --- [backend-academico] [           main]
b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 729 ms
2026-10-16T23:34:10.758-06:00  INFO 19520 --- [backend-academico] [           main]
m.e.backendacademico.config.CicloDeVida  : Bean inicializado en
2026-10-17T05:34:10.758253400Z
2026-10-16T23:34:11.003-06:00  INFO 19520 --- [backend-academico] [           main]
o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8080 (http) with context path '/'
2026-10-16T23:34:11.007-06:00  INFO 19520 --- [backend-academico] [           main]
m.e.b.BackendAcademicoApplication        : Started BackendAcademicoApplication in 1.514 seconds (process
running for 1.86)
```

```
2026-10-16T23:34:11.003-06:00  INFO 19520 --- [backend-academico] [
main] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port
8080 (http) with context path '/'
2026-10-16T23:34:11.007-06:00  INFO 19520 --- [backend-academico] [
main] m.e.b.BackendAcademicoApplication        : Started
BackendAcademicoApplication in 1.514 seconds (process running for 1.86)
2026-10-16T23:36:51.836-06:00  INFO 19520 --- [backend-academico]
[ionShutdownHook] o.s.boot.tomcat.GracefulShutdown        : Commencing
graceful shutdown. Waiting for active requests to complete
2026-10-16T23:36:51.848-06:00  INFO 19520 --- [backend-academico]
[tomcat-shutdown] o.s.boot.tomcat.GracefulShutdown        : Graceful
shutdown complete
2026-10-16T23:36:51.851-06:00  INFO 19520 --- [backend-academico]
[ionShutdownHook] m.e.backendacademico.config.CicloDeVida  :
Cierre del bean de demostración
[INFO] ------------------------------------------------------------------
------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------
------
[INFO] Total time:  02:44 min
[INFO] Finished at: 2026-10-16T23:36:51-06:00
[INFO] ------------------------------------------------------------------
------
¿Desea terminar el trabajo por lotes (S/N)? s
```





Reto




```java
package mx.edu.backendacademico.repository;

import static org.junit.jupiter.api.Assertions.assertEquals;
import static org.junit.jupiter.api.Assertions.assertTrue;

import java.util.concurrent.CountDownLatch;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.Future;
import java.util.concurrent.TimeUnit;
import java.util.concurrent.atomic.AtomicInteger;

import mx.edu.backendacademico.domain.Alumno;
import mx.edu.backendacademico.domain.EstatusAlumno;
import mx.edu.backendacademico.exception.MatriculaDuplicadaException;
import org.junit.jupiter.api.Test;

class AlumnoRepositoryConcurrenciaTest {

    @Test
    void dosTareasConLaMismaMatriculaProducenUnRegistroYUnConflicto()
            throws Exception {

        var repository = new AlumnoRepositoryEnMemoria();

        var preparadas = new CountDownLatch(2);
        var inicio = new CountDownLatch(1);
        var terminadas = new CountDownLatch(2);

        var registros = new AtomicInteger();
        var conflictos = new AtomicInteger();

        ExecutorService ejecutor = Executors.newFixedThreadPool(2);

        Runnable tarea = () -> {
            preparadas.countDown();

            try {
                inicio.await();

                repository.guardar(new Alumno(
```

```
                        null,
                        "A001",
                        "Alumno de prueba",
                        "prueba@uanl.mx",
                        EstatusAlumno.ACTIVO
                ));

                registros.incrementAndGet();

            } catch (MatriculaDuplicadaException e) {
                conflictos.incrementAndGet();

            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
                throw new RuntimeException(e);

            } finally {
                terminadas.countDown();
            }
        };

        try {
            Future<?> tarea1 = ejecutor.submit(tarea);
            Future<?> tarea2 = ejecutor.submit(tarea);

            assertTrue(preparadas.await(5, TimeUnit.SECONDS));

            inicio.countDown();

            assertTrue(terminadas.await(5, TimeUnit.SECONDS));

            tarea1.get();
            tarea2.get();

            assertEquals(1, registros.get());
            assertEquals(1, conflictos.get());
            assertEquals(1, repository.buscarTodos().size());

        } finally {
            inicio.countDown();
            ejecutor.shutdownNow();
        }
    }
}
```

```
PS C:\Users\andre\Downloads\Desarrollo-web-tecnolog-as-servidor\Unidades\Unidad
04\reto> .\mvnw.cmd "-Dtest=AlumnoRepositoryConcurrenciaTest" test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice
-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u04 0.0.4-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 12 source files with javac [debug parameters release 21] to
target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-
academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\Desarrollo-
web-tecnolog-as-servidor\Unidades\Unidad 04\reto\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico
---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 3 source files with javac [debug parameters release 21] to
target\test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider
org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running
mx.edu.backendacademico.repository.AlumnoRepositoryConcurrenciaTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.074 s --
in mx.edu.backendacademico.repository.AlumnoRepositoryConcurrenciaTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.370 s
[INFO] Finished at: 2026-10-16T23:56:31-06:00
[INFO] ------------------------------------------------------------------------
```